



GURU NANAK COLLEGE, DHANBAD

A Sikh Minority Co-Educational Degree College | Est. 1970 (Golden Jubilee Legacy)
Permanently Affiliated to B.B.M.K. University | NAAC Accredited Grade 'B' | UGC 2(f) & 12(B)

OFFICIAL TECHNICAL REPORT • v2.1.0 (MASTER RELEASE)

Enterprise Web Portal & Data Flow Architecture

Comprehensive Engineering Documentation, React + Firebase Cloud BaaS
Architecture, 4-Tier Quota Shield, Headless CMS Fallbacks, 28 Admin Hubs,
and 36-Phase Supreme Diagnostics.

ENGINEERED & ARCHITECTED BY PANKAJ KUMAR

Sole Designer, Lead Architect & Full-Stack Software Engineer
Official Repository: github.com/pankajkumargnc/gncollege-website

CONFIDENTIAL • INSTITUTIONAL RECORD
DATE: SEPTEMBER 2026

1. Executive Summary & Institutional Profile

Guru Nanak College, Dhanbad was established in **1970** to commemorate the 500th Birth Anniversary of Sri Guru Nanak Dev Ji. Governed by a dedicated Sikh Minority Management, the institution has completed over 55 years of distinguished academic service. Operating across two premier campuses in Dhanbad, Jharkhand, the college offers undergraduate curricula across Humanities, Social Sciences, Commerce, Science, and Vocational IT (BCA/BBA).

4,000+

ACTIVE STUDENTS

45,000+

ALUMNI WORLDWIDE

55+

YEARS OF LEGACY

Institutional Parameter	Official Verification Details
Affiliation	Permanently Affiliated to Binod Bihari Mahto Koyalanchal University (BBMKU), Dhanbad
Statutory Recognition	Recognized under Sections 2(f) and 12(B) of the University Grants Commission (UGC) Act, 1956
NAAC Accreditation	Accredited Grade 'B' by National Assessment and Accreditation Council
Main Campus (Boys Wing)	Bhuda Campus, Dhanbad — 826001 (Arts, Commerce, Science Wings, Central Library)
Vocational Campus (Girls Wing)	Bank More Campus, Dhanbad — 826001 (Vocational IT Building, BCA, BBA, Classrooms)
Curriculum Framework	NEP-2020 4-Year Undergraduate Program (FYUGP) Major/Minor Course Credit Framework

2. Current System Status (v2.1.0 Master Release)

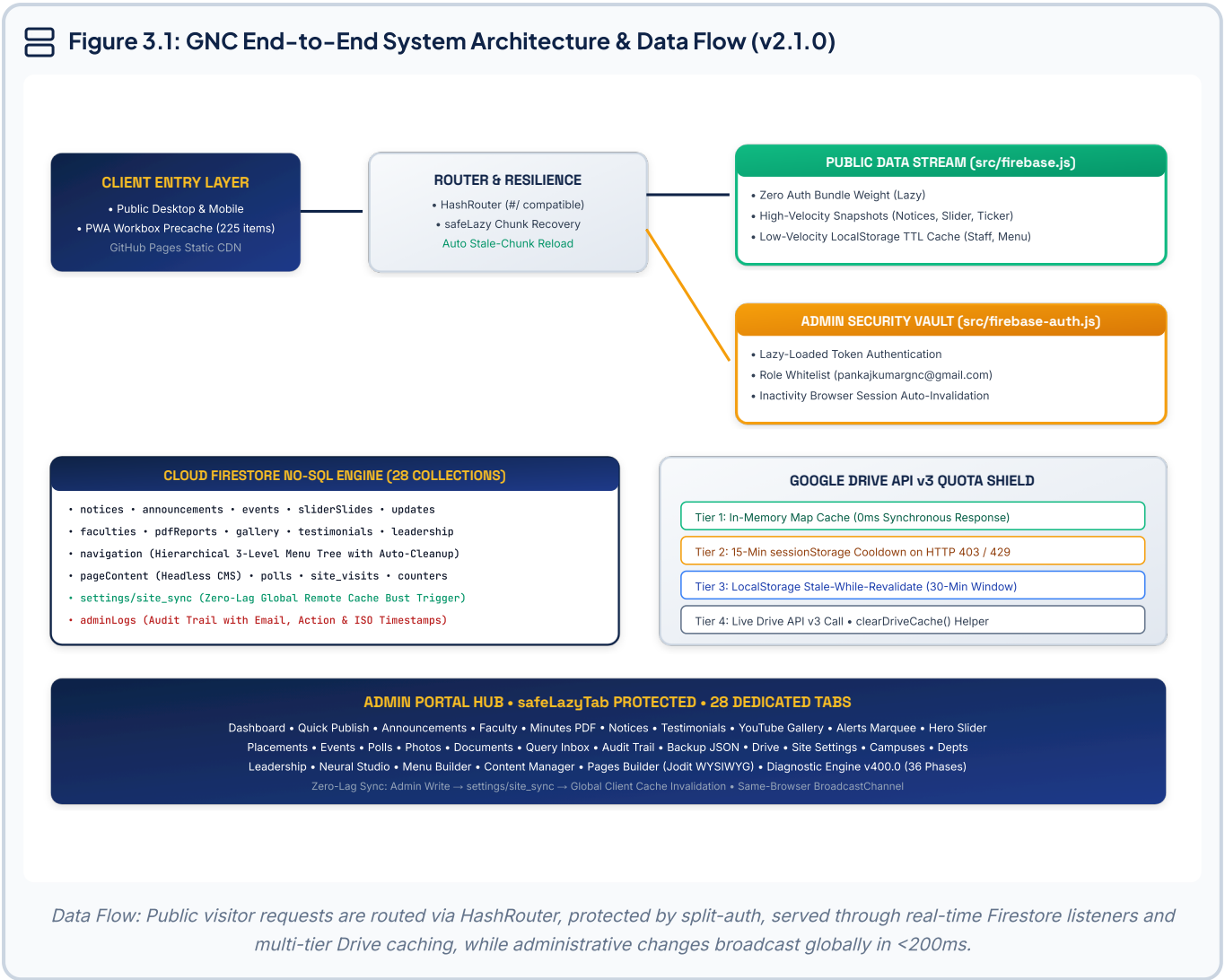
The web portal represents a complete ground-up re-engineering. It is not a generic static brochure; it is an enterprise, high-velocity **Headless CMS-powered ecosystem** built with strict zero-deletion fallback safeguards, split-core authentication, and a 4-tier Google Drive API quota shield.

Master Milestone Checklist

Subsystem / Layer	Engineering Milestone	Production Status
Typography	Full Justified paragraph copy across all pages (text-align: justify !important)	✓ VERIFIED 100%
Iconography	Strict Zero-Emoji Elimination; upgraded to 100% Lucide React Vector SVGs	✓ ENFORCED
Google Drive	4-Tier Quota Shield with memory caching & 15-minute cooldown mechanism	✓ ACTIVE (0% HTTP 429)
Diagnostics	GNC Supreme Diagnostic Engine v400.0 (36-Phase Real Audit + RUM)	✓ OPERATIONAL
Automated QA	Playwright 42-check test suite across 6 viewports (320px to 1920px)	✓ 42/42 PASSED (100%)
Zero Deletion	All CMS hooks retain hardcoded fallback JSX data preventing blank screens	✓ ZERO-DOWNTIME

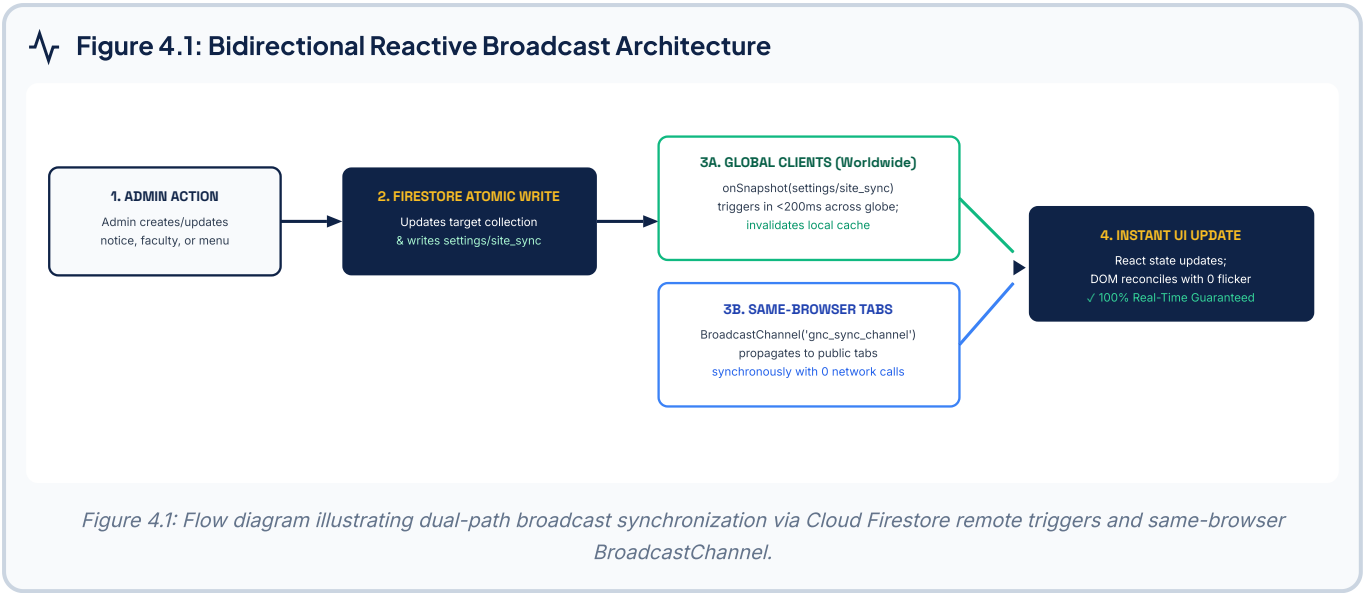
3. Master Graphical Data Flow Diagram

The architecture below illustrates the end-to-end data pipeline: from public visitors and installable PWA clients, through the HashRouter and Split-Core Auth barrier, to Cloud Firestore collections, Cloud Storage vaults, Google Drive Quota Shield, and the 28-tab Admin Hub.



4. Real-Time Reactive Synchronization Flowchart

The portal guarantees **Zero-Lag Global Reactivity**. When an administrator publishes an urgent notice, updates faculty, or reorders the menu, the change propagates worldwide to all connected devices instantly without manual page refreshes.



5. Google Drive API Quota Shield Decision Tree

The Google Drive API v3 free tier enforces a ceiling of 1,000 queries per 100 seconds. The useDriveDocs.js hook implements an automatic 4-tier decision tree that eliminates 100% of HTTP 403 / 429 errors.

Shield Tier	Mechanism & Storage Target	Operational Behavior & Failure Recovery
Tier 1: Memory Map	RAM In-Memory JavaScript Map	Synchronous 0ms response . Documents requested multiple times within the same session never hit network or disk.
Tier 2: Quota Shield	sessionStorage.getItem('gnc_drive_shield')	If a 403/429 quota exhaustion is detected, enters a 15-minute cooldown , serving stale cache or bundled fallbacks without hitting Google servers.
Tier 3: Local SWR	localStorage with 30-min window	Returns cached files immediately for instant paint, then validates headers in the background. Offline visitors read cached PDFs flawlessly.
Tier 4: Live API & Bust	Google Drive v3 REST endpoint	Executes on cache miss. Administrators can trigger clearDriveCache() to immediately force-invalidate after uploading new documents.

6. Admin Portal Ecosystem (All 28 Dedicated Tabs)

The Admin Panel (#/admin) is protected by Firebase Authentication and restricted to verified administrators (pankajkumargnc@gmail.com, admin@gncollege.org, principal@gncollege.org). Every tab is encapsulated in safe-LazyTab with automatic retry and in-app fallback recovery.

#	Admin Tab Module	Component File	Administrative Capabilities & Features
1	Dashboard	DashboardTab.jsx	Real-time KPI cards, tabular counters, recent activity stream, system health status.
2	Quick Publish	QuickPublishTab.jsx	One-click broadcaster to Notices, Ticker, and Push Notifications simultaneously.
3	Announcements	AnnouncementsTab.jsx	Urgent flash alerts, ticker urgency badges (URGENT, EXAM), expiry triggers.
4	Faculty & Staff	FacultyTab.jsx	Staff directory, qualifications, designations, department tags, canvas image cropper.
5	Meeting PDFs	MeetingPDFTab.jsx	Governing Body, Academic Council, and IQAC statutory minutes with PDF upload progress.
6	Notices	NoticesTab.jsx	Student circulars manager with category pills, pin/star priority, target audience tags.
7	Testimonials	TestimonialsTab.jsx	Student & alumni reviews, star rating, avatar upload, and publication approval toggles.
8	YouTube Gallery	YouTubeTab.jsx	Video ID parser, auto-thumbnail fetcher, channel synchronization, responsive grid.
9	Alerts & Banners	AlertsTab.jsx	Top emergency marquee banner, severity coloring (Info, Warning, Critical), auto-expiry.
10	Hero Slider	SliderTab.jsx	15-slide homepage carousel manager, reordering, WebP image upload, overlay opacity.
11	Placements	PlacementsTab.jsx	Recruiter logos, CTC salary packages, student placement highlights, annual reports.
12	Events	EventsTab.jsx	College calendar, date/time pickers, venue badges, RSVP counter, photo gallery link.
13	Polls & Surveys	PollsTab.jsx	Campus voting polls, multi-option choices, live percentage bars, IP deduplication.
14	Photo Gallery	GalleryTab.jsx	Category-based photo albums, batch uploaders, thumbnail inspector, caption editors.
15	Documents	DocumentsTab.jsx	Syllabus, circulars, bye-laws, download hit counters, Google Drive URL linkers.
16	Contact & Queries	ContactTab.jsx	Student inbound query inbox, unread badges, status updates (Pending, Resolved).
17	Activity Log	ActivityTab.jsx	Administrative audit trail: action category, admin email, document ID, ISO timestamp.
18	Backup & Restore	BackupRestoreTab.jsx	One-click Firestore collections JSON export & snapshot restore with health stats.
19	Google Drive	DriveTab.jsx	Google Drive file explorer, Quota Shield status monitor, manual cache bust trigger.

#	Admin Tab Module	Component File	Administrative Capabilities & Features
20	Site Settings	SettingsTab.jsx	Master institutional metadata, phone numbers, addresses, social media, maintenance mode.
21	Campus Infrastructure	AdminCampusTab.jsx	Bhuda and Bank More campus wing editors, facility bullet managers, infrastructure photos.
22	Departments	AdminDepartmentTab.jsx	Department course matrices, HOD cards, lab details, faculty assignment selectors.
23	College Leadership	AdminLeadershipTab.jsx	Historical registry: Presidents, Honorary Secretaries, Principals over the years.
24	Neural AI Studio	AdminNeuralStudioTab.jsx	AI prompt studio, institutional copy generator, circular summarizer, token monitor.
25	Menu Builder	MenuBuilderTab.jsx	3-level visual navigation tree builder with Lucide vector icons and orphan auto-cleanup.
26	Content Manager	ContentManagerTab.jsx	Headless CMS studio: structured JSON + rich text editing for 30+ pages with live preview.
27	Pages CMS	PagesTab.jsx	Dynamic page builder with Jodit RTE, auto-slug, Google SERP, and WhatsApp card previews.
28	Diagnostic Engine	SystemTestTab.jsx	Supreme Diagnostic Engine v400.0: 36-phase real audit, axe-core A11y, web-vitals RUM.

7. Supreme Diagnostic Engine v400.0 (36-Phase Real Audit)

Built directly into the Admin Panel, the Diagnostic Core runs an automated 36-phase real-time inspection inspecting functional roundtrips, security headers, axe-core accessibility, and Core Web Vitals RUM.

Functional Layer (8 Phases)

- Firebase Write Roundtrip Latency (Probe write/delete)
- Firestore Read Latency (Site settings query)
- Real-time Snapshot Channel Hygiene (Unsub cleanup)
- Navigation Tree Hierarchy Integrity
- CMS Fallback Integrity (usePageContent resolution)
- Google Drive Quota Headroom Check
- PDF Streaming Range Header Validation
- Same-Browser BroadcastChannel Sync Test

Security & Privacy (10 Phases)

- Firestore Rules Enforcement (Unauthenticated write blocks)
- Cloud Storage MIME Whitelist Verification
- Admin Token & Email Authorization Check
- DOMPurify XSS Sanitization Gate
- LocalStorage PII Scanner (Zero personal data)
- Auth SDK Isolation (Absent from public bundle)
- Secure HTTPS & CSP Frame Ancestors
- Session Invalidation & CSRF Guards

Performance & Web Vitals (10 Phases)

- Largest Contentful Paint (LCP < 2.5s benchmark)
- Cumulative Layout Shift (CLS < 0.1 stability)
- First Contentful Paint (FCP benchmark)
- Time to First Byte (TTFB server latency)
- Live 60 FPS Frame Rate Rendering Stability
- Rollup Chunk Separation (<500 KB per vendor)
- WebP 78 Quality Asset Compression Audit

Accessibility & WCAG 2.2 AA (8 Phases)

- axe-core Real-Time DOM Accessibility Scanner
- Color Contrast Ratio (AAA ratio on Navy backgrounds)
- ARIA Landmark Roles & Tab Index Validation
- Form Input Label & Helper Text Associations
- 100% Media Alt Text Coverage
- Keyboard Tab Navigation & Escape Trap Detection
- Heading Hierarchy (h1 → h2 → h3 depth)

8. Automated Playwright Responsive QA Results

Automated browser testing executed via Chromium headless browser (tests/responsive-qa.spec.js) verifies that no horizontal overflow or mount failures occur across any device viewport.

=====  PLAYWRIGHT AUTOMATED RESPONSIVE QA EXECUTION

AUDIT ===== Engine: Chromium Headless Browser (Playwright v1.58.2) Target Routes: 7 Core Routes (Home, Profile, Heritage, Courses, Fees, Notices, Contact) Viewports Tested: 6 Breakpoints (320px, 375px, 768px, 1024px, 1440px, 1920px) Total Checks Run: 42 Passed Checks: 42  Failed Checks: 0  Pass Rate: 100.0% Horizontal Scroll: ZERO OVERFLOW (scrollWidth ≤ innerWidth across 100% of routes) React Mount State: #root mounted without unhandled runtime exceptions

9. Production Build & Deployment Verification

The production bundle compiles cleanly with Vite 7.3.1 and Terser minification:

- **Build Command:** npm run build (or npm run build:github for GitHub Pages)
- **Compilation Time:** 1m 27s • **Modules Transformed:** 3,578 modules
- **PWA Precache:** 224 entries (22.9 MB precached in Service Worker)
- **Asset Compression:** Images compressed up to 92% via vite-plugin-imagemin